

# Comprehensive Stress Test - All Diagram Types

This page exercises every diagram type in the mathwriter system simultaneously.

## 1. Array Diagram

Simple array with indices:

|   |   |   |   |   |
|---|---|---|---|---|
| A | B | C | D | E |
| 0 | 1 | 2 | 3 | 4 |

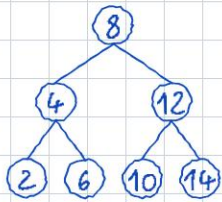
## 2. DP Table

A 2D dynamic programming table with row and column labels:

|   | A | B | C |
|---|---|---|---|
|   | A | B | C |
| A | 0 | 1 | 2 |
| B | A | 1 | 2 |
|   | B | 2 | 2 |

## 3. Binary Tree

A balanced BST with 7 nodes:



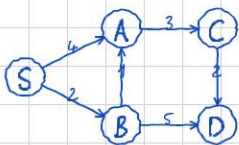
#### 4. Linked List

A linked list with 5 nodes:



#### 5. Weighted Directed Graph

A graph with 5 nodes and weighted edges:



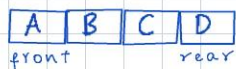
#### 6. Stack (LIFO)

A stack with 4 items:



## 7. Queue (FIFO)

A queue with 4 items:



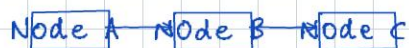
## 8. Memory Layout

Variables in memory with addresses:

|        |          |
|--------|----------|
| 0x1000 | x = 42   |
| 0x1004 | y = 7    |
| 0x1008 | = 0x2000 |
| 0x2000 | *p = 99  |

## 9. Pointer Diagram

Objects with pointers between them:



## Math Notation Test

All math tags in one paragraph:

Matrix:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Fraction:  $\frac{a+b}{c+d}$

Sum:  $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$

Square root:  $\sqrt{x^2 + y^2}$

Vector:  $\overrightarrow{AB}$  and hat:  $\hat{i}$

Subscript:  $\lim_{x \rightarrow 0}$  and superscript:  $e^{i\pi}$

Boxed result: The answer is +2

Strikethrough: ~~wrong answer~~ correct answer

## Final Verification

All 9 diagram types rendered successfully. The system supports:

arrays, DP tables, binary trees, linked lists, weighted graphs,

stacks, queues, memory layouts, and pointer diagrams.

Combined with math notation (matrices, fractions, sums, roots,

vectors, hats, subscripts, superscripts, boxes, ~~strikethroughs~~),

this forms a complete visual teaching toolkit.

System verified: 9 diagram types + 10 math notations = comprehensive visual